Synoptic Question 22: The Physics of a Commercial Airliner

Reading Passage

Read the following article and then answer the questions that follow.

A modern commercial airliner, such as a Boeing 787, is a marvel of applied physics. Its flight can be broken down into several phases: take-off, climb, cruise, and landing. Each phase is governed by a balance of four primary forces: **Lift** (upward), **Weight** (downward), **Thrust** (forward), and **Drag** (backward).

Lift is generated primarily by the wings, which are shaped as aerofoils. This shape forces the air flowing over the top surface to travel faster than the air flowing underneath. According to Bernoulli's principle, this creates a region of lower pressure above the wing, resulting in a net upward force.

Thrust is provided by high-bypass turbofan engines. These engines work by pulling in a large volume of air with a massive fan. A small portion of this air is compressed, mixed with fuel, and ignited in a combustion chamber. This high-energy exhaust is expelled at high speed, but the majority of the thrust (around 80%) comes from accelerating the large volume of "bypassed" air that just passes through the fan. This entire process is a direct application of Newton's third law and the conservation of momentum.

At a cruising altitude of over 10 km, the outside air pressure and temperature are extremely low. The aircraft's cabin must be pressurised to maintain a safe environment for passengers, using compressed air bled from the engines.

Data:

Data for a typical Boeing 787 in operation:

Table 1: Data for Airliner

Parameter	**Value**
Maximum take-off mass, m_{max}	254,000 kg
Total wing area, A_{wing}	360 m^2
Length of take-off runway, d	2800 m
Take-off speed, v_{to}	280 km/h
Cruising speed (constant), v_{cruise}	900 km/h
Cruising altitude, h	12,000 m
Mass flow rate of air (per engine), $\dot{m}$	1100 kg s^{-1}
Exhaust gas velocity (relative to engine), v_{ex}	310 m s^{-1}
Volume of pressurised cabin, V_{cabin}	680 m^3
Pressure in cabin, P_{cabin}	7.50×10^4 Pa (8000 ft equivalent)
Temperature in cabin, T_{cabin}	22.0 °C
Molar mass of air, M_{air}	0.029 kg mol^{-1}
Ideal gas constant, R	8.31 J K^{-1} mol^{-1}

Questions

(a) The aircraft, at its maximum mass, accelerates from rest for take-off.

  (i) Convert the take-off speed of 280 km/h to m s^{-1}.

  (ii) Assuming constant acceleration, calculate the acceleration a required for take-off.

  (iii) Calculate the work done on the aircraft to reach its take-off speed.

(b) The aircraft is now at its cruising altitude and speed.

  (i) Calculate the kinetic energy of the aircraft at its cruising speed.

  (ii) Calculate the gain in gravitational potential energy (GPE) of the aircraft from sea level to its cruising altitude.

(c) While cruising, the aircraft maintains a constant velocity and altitude.

  (i) State the magnitude of the net force acting on the aircraft.

(ii) Calculate the total upward lift force required.

(iii) The lift force is generated by the pressure difference above and below the wings. Using your answer from (c)(ii), calculate the average pressure difference (ΔP) between the top and bottom surfaces of the wings.

(d) The turbofan engines provide the thrust. The thrust F is given by $F = \dot{m}(v_{out} - v_{in})$. Assume the engines are tested on the ground ($v_{in} = 0$) and the air is expelled at v_{ex}.

(i) State the physics principle that explains how the engine generates thrust.

(ii) Calculate the thrust produced by a single engine.

(e) The aircraft cabin is pressurised.

(i) Convert the cabin temperature of 22.0 °C to Kelvin.

(ii) Assuming the air in the cabin behaves as an ideal gas, calculate the total mass of air inside the pressurised cabin.

(f) An observer on the ground hears the sound from the engines as the aircraft passes directly overhead at its cruising speed. The engines emit a sound with a primary frequency of 150 Hz. (Speed of sound at sea level = 340 m s^{-1}).

(i) State the name of the phenomenon that causes the observer to hear a different frequency from the source.

(ii) Calculate the frequency the observer hears as the aircraft moves **away** from them.

Solutions

(a) (i) $v_{to} = 280 \text{ km/h} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 77.78 \text{ m s}^{-1}$.

(ii) $v^2 = u^2 + 2as \implies a = \frac{v^2 - u^2}{2s}$
$a = \frac{(77.78)^2 - 0}{2(2800)} = 1.079 \text{ m s}^{-2} \approx 1.08 \text{ m s}^{-2}$.

(iii) Work Done $= \Delta KE = \frac{1}{2}mv_{to}^2$
Work Done $= \frac{1}{2}(254000)(77.78)^2 = 7.68 \times 10^8$ J.

(b) (i) $v_{cruise} = 900 \text{ km/h} \times \frac{1000}{3600} = 250 \text{ m s}^{-1}$.
$KE = \frac{1}{2}mv_{cruise}^2 = \frac{1}{2}(254000)(250)^2 = 7.94 \times 10^9$ J.

(ii) $GPE = mgh = (254000)(9.81)(12000) = 2.99 \times 10^{10}$ J.

(c) (i) The velocity and altitude are constant, so acceleration is zero. By Newton's First Law, the net force is **zero**.

(ii) For the vertical forces to be balanced, Lift = Weight.
$Lift = W = mg = (254000)(9.81) = 2.49 \times 10^6$ N.

(iii) $Force = \text{Pressure} \times \text{Area} \implies F_{lift} = \Delta P \times A_{wing}$
$\Delta P = \frac{F_{lift}}{A_{wing}} = \frac{2.49 \times 10^6}{360} = 6917 \text{ Pa} \approx 6920 \text{ Pa}$.

(d) (i) **Conservation of Linear Momentum** (or Newton's Third Law). The engine exerts a force on the air, pushing it backward. By N3L, the air exerts an equal and opposite (forward) force on the engine.

(ii) $F = \dot{m}(v_{out} - v_{in}) = (1100)(310 - 0) = 341000$ N (or 341 kN).

(e) (i) $T(K) = T(°\text{C}) + 273.15 = 22.0 + 273.15 = 295.15$ K.

(ii) Using Ideal Gas Law: $PV = nRT \implies n = \frac{PV}{RT}$
$n = \frac{(7.50 \times 10^4)(680)}{(8.31)(295.15)} = 20800$ mol.
Mass m = moles × Molar mass $= n \times M_{air}$
$m = (20800)(0.029) = 603$ kg.

(f) (i) The **Doppler effect**.

(ii) The source is moving away from the stationary observer.
$f' = f\left(\frac{v_{sound}}{v_{sound} + v_{source}}\right)$
$v_{source} = v_{cruise} = 250 \text{ m s}^{-1}$.
$f' = 150\left(\frac{340}{340+250}\right) = 150\left(\frac{340}{590}\right) = 86.4$ Hz.

Cambridge IGCSE Physics (0625) – Paper 4 Style

Synoptic Question 22: Commercial Airliner Physics

Variant 2 (Higher Difficulty)

Instructions (CIE P4 style)

- Answer all questions.
- Show all working clearly and give appropriate units.
- Use $g = 9.81\,\mathrm{m\,s^{-2}}$ unless stated otherwise.
- Calculators may be used.

Data for a typical Boeing 787 in operation

Parameter	Value
Maximum take-off mass, m_{max}	$2.54 \times 10^5\,\mathrm{kg}$
Total wing area, A_{wing}	$3.60 \times 10^2\,\mathrm{m^2}$
Length of take-off runway, d	$2.800 \times 10^3\,\mathrm{m}$
Take-off speed, v_{to}	$2.80 \times 10^2\,\mathrm{km\,h^{-1}}$
Cruising speed, v_{cruise}	$9.00 \times 10^2\,\mathrm{km\,h^{-1}}$
Cruising altitude, h	$1.2000 \times 10^4\,\mathrm{m}$
Mass flow rate of air per engine, $\dot{m}$	$1.100 \times 10^3\,\mathrm{kg\,s^{-1}}$
Exhaust speed (relative to engine), v_{ex}	$3.10 \times 10^2\,\mathrm{m\,s^{-1}}$
Cabin volume, V_{cabin}	$6.80 \times 10^2\,\mathrm{m^3}$
Cabin pressure, P_{cabin}	$7.50 \times 10^4\,\mathrm{Pa}$
Cabin temperature	$2.20 \times 10^1\,^\circ\mathrm{C}$
Molar mass of air, M_{air}	$2.9 \times 10^{-2}\,\mathrm{kg\,mol^{-1}}$
Ideal gas constant, R	$8.31\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Speed of sound at sea level	$3.40 \times 10^2\,\mathrm{m\,s^{-1}}$

Additional data for this variant

- Total engine thrust available at take-off (both engines combined) $= 6.82 \times 10^5\,\mathrm{N}$.
- During a steady climb at cruise speed, drag is $1.20 \times 10^5\,\mathrm{N}$ and average climb angle is small.
- Jet fuel energy density $= 4.3 \times 10^7\,\mathrm{J\,kg^{-1}}$.

Question 1

(a) Convert v_{to} and v_{cruise} into $\mathrm{m\,s^{-1}}$.

(b) Assuming constant acceleration from rest over the runway distance, calculate:

  (i) the take-off acceleration,

  (ii) the time taken to reach take-off speed.

(c) Calculate the average resultant horizontal force needed for this acceleration at maximum take-off mass.

(d) The total engine thrust at take-off is $6.82 \times 10^5\,\mathrm{N}$. Determine the average resistive force (drag + rolling resistance) during the take-off run.

Question 2

(a) Calculate the aircraft kinetic energy at cruising speed.

(b) Calculate the gain in gravitational potential energy from sea level to cruise altitude.

(c) Deduce the minimum mechanical work needed to bring the aircraft from rest at sea level to cruise conditions (ignore losses).

(d) The climb phase takes 1.8×10^1 min. Estimate the minimum average power output required for this energy increase.

Question 3

(a) At steady, level cruise, state the net force on the aircraft and explain why.

(b) Calculate the lift force required at cruise.

(c) Use your answer to find the average pressure difference ΔP between lower and upper wing surfaces.

(d) Open-ended concept question:
A student says: *"Lift is only caused by faster air over the top surface, so Newton's laws are not relevant."*
Discuss this statement using appropriate physics ideas, and explain why a complete explanation of lift should include more than one model.

Question 4

(a) Using $F = \dot{m}(v_{\text{out}} - v_{\text{in}})$ with ground-test $v_{\text{in}} = 0$, calculate the thrust from one engine.

(b) Compare your result with the stated total take-off thrust and comment briefly.

(c) Open-ended design question:
Commercial engines use very high bypass ratios. Explain, with reference to momentum change, efficiency and noise, why accelerating a large mass of air by a smaller speed change is usually preferred to accelerating a small mass by a very large speed change.

Question 5

(a) Convert cabin temperature to kelvin and calculate the amount of substance n of air in the cabin.

(b) Hence calculate the total mass of air in the cabin.

(c) The aircraft climbs to 1.2000×10^4 m where external pressure is much lower than cabin pressure. Explain two engineering challenges this pressure difference causes and one method used to manage each challenge.

Question 6

(a) An observer hears engine sound of emitted frequency 1.50×10^2 Hz when the aircraft moves directly away at cruise speed. Calculate the observed frequency.

(b) Open-ended evaluation:
Explain why the answer from (a) is an idealised value for a real flyover. Discuss at least three physical factors that can cause differences between calculated and observed frequency.

Sample Solution – Variant 2

Question 1

(a) $v_{\text{to}} = 280 \times \frac{1000}{3600} = 7.78 \times 10^1\,\text{m}\,\text{s}^{-1}$; $\quad v_{\text{cruise}} = 900 \times \frac{1000}{3600} = 2.50 \times 10^2\,\text{m}\,\text{s}^{-1}$.
(b)(i) $a = \frac{v^2-u^2}{2d} = \frac{(77.8)^2}{2(2800)} = 1.08\,\text{m}\,\text{s}^{-2}$.
(b)(ii) $t = \frac{v-u}{a} = \frac{77.8}{1.08} = 7.22 \times 10^1\,\text{s}$.
(c) $F_{\text{resultant}} = ma = (2.54 \times 10^5)(1.08) = 2.74 \times 10^5\,\text{N}$.
(d) $F_{\text{resistive}} = F_{\text{thrust,total}} - F_{\text{resultant}} = 6.82 \times 10^5 - 2.74 \times 10^5 = 4.08 \times 10^5\,\text{N}$.

Question 2

(a) $E_k = \frac{1}{2}mv^2 = \frac{1}{2}(2.54 \times 10^5)(250)^2 = 7.94 \times 10^9\,\text{J}$.
(b) $\Delta E_p = mgh = (2.54 \times 10^5)(9.81)(12000) = 2.99 \times 10^{10}\,\text{J}$.
(c) $W_{\min} = E_k + \Delta E_p = 3.78 \times 10^{10}\,\text{J}$.
(d) $t = 18 \times 60 = 1.080 \times 10^3\,\text{s}$, so $P_{\min} = \frac{W_{\min}}{t} = \frac{3.78\times10^{10}}{1080} = 3.50 \times 10^7\,\text{W}$ (about $3.5 \times 10^1\,\text{MW}$).

Question 3

(a) Net force is zero because velocity is constant (no acceleration) in both vertical and horizontal directions in steady level cruise.
(b) $L = W = mg = (2.54 \times 10^5)(9.81) = 2.49 \times 10^6\,\text{N}$.
(c) $\Delta P = \frac{L}{A} = \frac{2.49\times10^6}{360} = 6.92 \times 10^3\,\text{Pa}$.
(d) A complete response should include that Bernoulli pressure differences are useful, but lift also requires downward momentum change of air (Newton's third law), angle of attack, circulation and viscous flow effects.

Question 4

(a) $F = \dot{m}(v_{\text{out}} - v_{\text{in}}) = 1100(310 - 0) = 3.41 \times 10^5\,\text{N}$ per engine.
(b) Two engines give $2 \times 3.41 \times 10^5 = 6.82 \times 10^5\,\text{N}$, consistent with stated take-off thrust.
(c) High bypass engines improve propulsive efficiency because a larger air mass gets a smaller velocity increase, reducing kinetic energy wasted in the exhaust and often reducing noise.

Question 5

(a) $T = 22.0 + 273.15 = 2.9515 \times 10^2\,\text{K}$. $n = \frac{PV}{RT} = \frac{(7.50\times10^4)(680)}{(8.31)(295.15)} = 2.08 \times 10^4\,\text{mol}$.
(b) $m = nM = (2.08 \times 10^4)(0.029) = 6.03 \times 10^2\,\text{kg}$.
(c) Typical challenges: fuselage stress due to pressure differential (managed by strong pressure vessel and inspection for fatigue); maintaining breathable cabin air and temperature (managed by controlled bleed-air/compressor systems and outflow valves).

Question 6

(a) Doppler shift for source moving away: $f' = f\left(\frac{v}{v+v_s}\right) = 150\left(\frac{340}{340+250}\right) = 8.64 \times 10^1\,\text{Hz}$.
(b) Real values differ because speed/heading vary during flyover, wind alters effective sound

speed, temperature gradients refract sound, engine frequency spectrum is broad and changing with throttle, and observer position is rarely exactly on-axis.

Mark Scheme – Variant 2 (CIE P4 style)

Total: 50 marks

Q1 [8]

(a) $v_{\text{to}} = 77.8\ \text{m s}^{-1}$ and $v_{\text{cruise}} = 250\ \text{m s}^{-1}$ (1+1) [2]
(b)(i) Correct use of kinematics and $a = 1.08\ \text{m s}^{-2}$ [2]
(b)(ii) $t = 72$ s (allow 71–73 s) [1]
(c) $F = ma$ and 2.74×10^5 N [2]
(d) Resistive force $= 4.08 \times 10^5$ N [1]

Q2 [8]

(a) KE $= 7.94 \times 10^9$ J [2]
(b) GPE gain $= 2.99 \times 10^{10}$ J [2]
(c) Total minimum work $= 3.78 \times 10^{10}$ J [2]
(d) Average minimum power $= 3.5 \times 10^7$ W (35 MW) [2]

Q3 [9]

(a) Net force zero with clear link to constant velocity/no acceleration [2]
(b) Lift equals weight with correct value 2.49×10^6 N [2]
(c) Pressure difference $\Delta P \approx 6.9 \times 10^3$ Pa [2]
(d) Open-ended explanation [3]:

1 Bernoulli description correctly stated.

1 Newton's third law / downward momentum transfer stated.

1 Limitation/extension (angle of attack, circulation, viscosity, stall behaviour, or need for multiple models).

Q4 [8]

(a) Single-engine thrust 3.41×10^5 N [2]
(b) Comparison with total thrust and consistency comment [2]
(c) Open-ended design discussion [4]:

1 Momentum argument: large mass flow with smaller Δv.

1 Higher propulsive efficiency / less wasted exhaust KE.

1 Noise reduction / lower jet velocity benefits.

1 Coherent physics reasoning and correct terminology.

Q5 [8]

(a) Temperature in K and correct n [3]
(b) Correct cabin air mass [1]
(c) Open-ended engineering response [4]:

1+1 Two valid challenges (e.g. structural fatigue, sealing, environmental control, hypoxia risk).

1+1 Appropriate mitigation for each challenge.

Q6 [9]

(a) Correct Doppler identification/formula and $f' \approx 86$ Hz [3]
(b) Open-ended evaluation [6]: award up to 6 for well-developed points, e.g.

- aircraft speed and direction not constant,
- wind and atmospheric motion,
- temperature/pressure gradients and refraction,
- distributed and changing engine frequencies,
- geometry of observer position,
- reflections/ground effects/background noise.

Examiner note: For open-ended parts, award marks for valid physics, clear reasoning, and coherent communication even if wording differs from sample solutions.